Power and impedance relations

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Input impedance of the dissipation less line

$$E = \frac{E_R}{2}\left(\frac{Z_R+R_0}{Z_R}\right)\left[e^{j\beta s} + K\, e^{-j\beta s}\right]$$

$$I = \frac{I_R}{2}\left(\frac{Z_R+R_0}{R_0}\right)\left[e^{j\beta s} - K\, e^{-j\beta s}\right]$$

$$Z_s = \frac{E}{I} = R_0\frac{\left[e^{j\beta s} + K\, e^{-j\beta s}\right]}{\left[e^{j\beta s} - K\, e^{-j\beta s}\right]}$$

But, $K = \frac{Z_R - R_0}{Z_R + R_0} = |K|\, e^{j\varphi}$

$$Z_s = \frac{E}{I} = R_0\frac{\left[e^{j\beta s} + |K|\, e^{-j(\beta s-\varphi)}\right]}{\left[e^{j\beta s} - |K|\, e^{-j(\beta s-\varphi)}\right]} = R_0\frac{1\angle\beta s + |K|\angle\phi-\beta s}{1\angle\beta s - |K|\angle\phi-\beta s}$$

Divide both numerator and denominator by $1\angle\beta s$ we get

$$Z_s = R_0\frac{1+|K|\angle\phi-2\beta s}{1-|K|\angle\phi-2\beta s}$$

$$\frac{Z_s}{R_0} = \frac{1 + |K| \angle \phi - 2\beta s}{1 - |K| \angle \phi - 2\beta s}$$

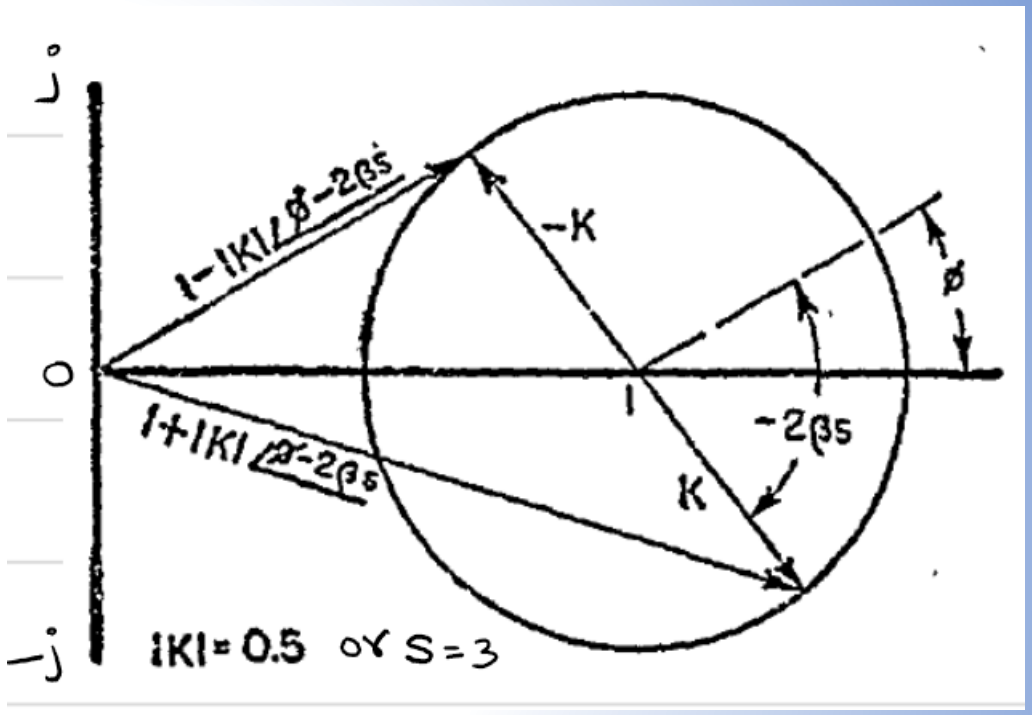
When ϕ - 2βs = 0 we will have impedance as a pure resistance

We will have a maximum resistance

When ϕ - 2βs = 0 or 2nπ

$$\frac{Z_s}{R_0} = \frac{1 + |\mathrm{K}|}{1 - |\mathrm{K}|} = \mathrm{S}$$

$R_{max} = SR_0$



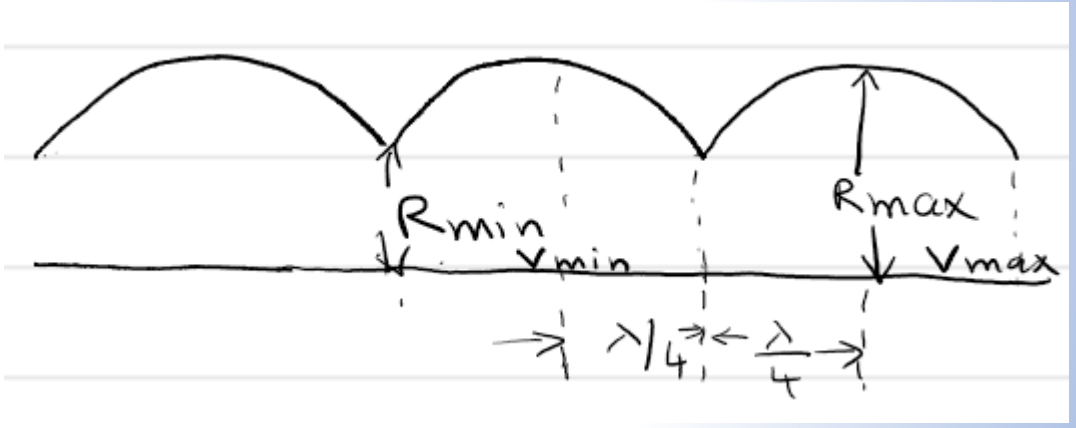
when $\phi - 2\beta s = \pi$ or $(2n+1)\pi$

$$\frac{Z_s}{R_0} = \frac{1-|K|}{1+|K|} = \frac{1}{S}$$

$$R_{min} = \frac{R_0}{S}$$

$$R_{max} \cdot R_{min} = R_0^2$$

$$R_0 = \sqrt{R_{max} \cdot R_{min}}$$



Note: Physical properties of RF line repeats after λ/2 interval

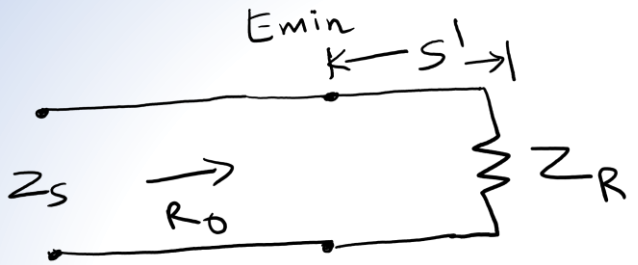
Power and impedance measurement on lines:

$$P = \frac{E_{max}^2}{R_{max}} = \frac{E_{min}^2}{R_{min}}$$

$$P^2 = \frac{|E_{max}^2|\,|E_{min}^2|}{R_0 \cdot R_0}$$

$$P = \frac{|E_{max}|\,|E_{min}|}{R_0} = (|I_{max}|\,|I_{min}|)\,R_0$$

To find unknown load impedance:



$$Z_S = R_0 \frac{Z_R + j R_0 \tan\beta s}{R_0 + j Z_R \tan\beta s}$$

At $s = s'$, E_{min} occurs and $R_{min} = \frac{R_0}{S}$

$$\frac{R_0}{S} = R_0 \frac{Z_R + j R_0 \tan\beta s'}{R_0 + j Z_R \tan\beta s'}$$

$$Z_R(S - j\tan\beta s') = R_0(1 - jS\tan\beta s') \Rightarrow \boxed{Z_R = R_0 \frac{(1 - jS\tan\beta s')}{S - j\tan\beta s'}}$$

(4) # In an experiment it is found that the first voltage minima occurs at a distance 0.085λ from the load. If $S = 3.5$, calculate the unknown load impedance. The operating frequency is 300 MHz and $R_0 = 300\Omega$.

$$\therefore Z_R = \frac{300(1 - j\,3.5 \tan \frac{2\pi}{\lambda} \cdot 0.085\lambda)}{(3.5 - j \tan \frac{2\pi}{\lambda} \cdot 0.085\lambda)}$$

$$= \frac{300 - j\,620.96}{3.5 - j\,0.591}$$

$$= \frac{689.63 \angle -1.12}{3.54 \angle 0.16} = 194.8 \angle -0.96$$

$$\frac{689.63 \angle -64.213}{3.549 \angle -9.58}$$

$$= 194.3 \angle 54.63$$

Reflection losses on the unmatched line:

$$E = \frac{E_R}{2}\left(\frac{Z_R+R_0}{Z_R}\right)\left[e^{j\beta s} + |K|\, e^{-j\beta s}.e^{j\varphi}\right]$$

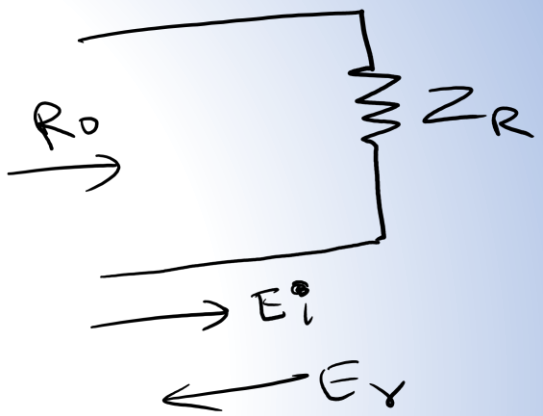
$$I = \frac{I_R}{2}\left(\frac{Z_R+R_0}{R_0}\right)\left[e^{j\beta s} - |K|\, e^{-j\beta s}.e^{j\varphi}\right]$$

$$|E_{max}| = |E_i| + |E_R|$$

$$E = \frac{E_R}{2}\left(\frac{Z_R+R_0}{Z_R}\right)\left[\,1 + |K|\,\right]$$

$$|E_{min}| = |E_i| - |E_R|$$

$$E = \frac{E_R}{2}\left(\frac{Z_R+R_0}{Z_R}\right)\left[\,1 - |K|\,\right]$$



$$S = \frac{|E_{max}|}{|E_{min}|} = \frac{|E_i| + |Er|}{|E_i| - |Er|} = \frac{1 + |K|}{1 - |K|}$$

Power delivered to the load

$$P = \frac{|E_{max}| . |E_{min}|}{R_0}$$

$$= \frac{\{|Ei| + |Er|\} . \{|Ei| - |Er|\}}{R_0}$$

$$= \frac{|E_i^2|}{R_0} - \frac{|Er^2|}{R_0}$$

P = Pi – Pr

= Incident power – reflected power

- The ratio of power delivered to the load to the power transmitted by the incident wave is

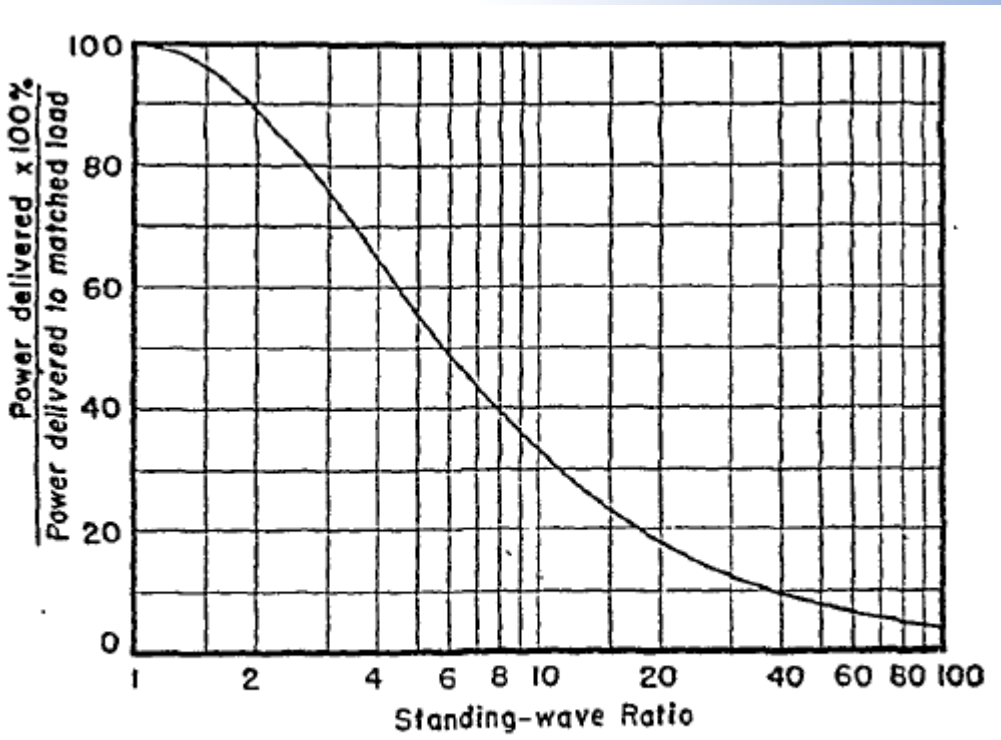
$$\frac{P}{pi} = \frac{Pi - Pr}{pi} = \frac{|E_i^2| - |E_r^2|}{|E_i^2|}$$

$$= 1 - \frac{|E_r^2|}{|E_i^2|}$$

$$= 1 - |K|^2$$

$$= 1 - \left(\frac{S-1}{S+1}\right)^2 = \frac{4S}{(S+1)^2}$$

$$\boxed{P = \frac{4S}{(S+1)^2} P_i}$$



Reflection losses as a function of standing-wave ratio.


Thank
you